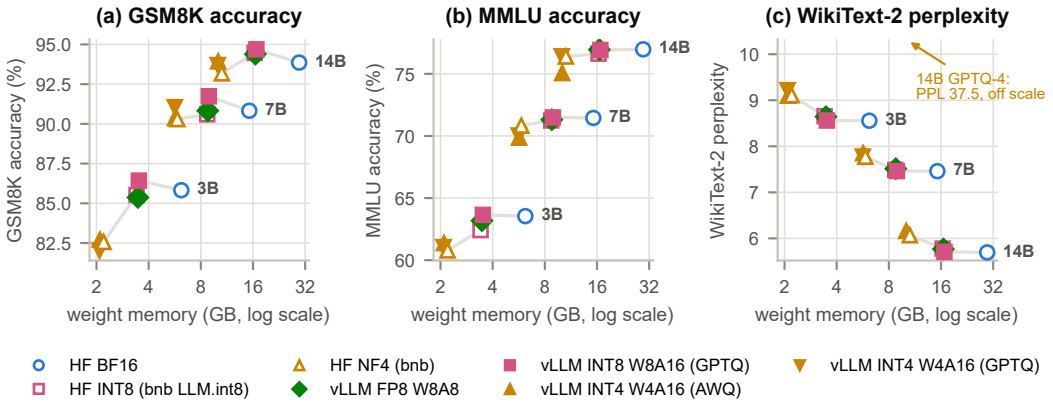

(a) GSM8K accuracy
GSM8K accuracy (%)
95.0
92.5
90.0
87.5
85.0
82.5
14B
7B
3B
2
4
8
16
32
weight memory (GB, log scale)
(b) MMLU accuracy
MMLU accuracy (%)
75
70
65
60
14B
7B
3B
2
4
8
16
32
weight memory (GB, log scale)
(c) WikiText-2 perplexity
WikiText-2 perplexity
10
9
8
7
6
14B GPTQ-4:
PPL 37.5, off scale
3B
7B
14B
2
4
8
16
32
weight memory (GB, log scale)
HF BF16
HF INT8 (bnb LLM.int8)
HF NF4 (bnb)
vLLM FP8 W8A8
vLLM INT8 W8A16 (GPTQ)
vLLM INT4 W4A16 (AWQ)
vLLM INT4 W4A16 (GPTQ)